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word2vec, RNNs, and Transformers

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Evolution of Language Models

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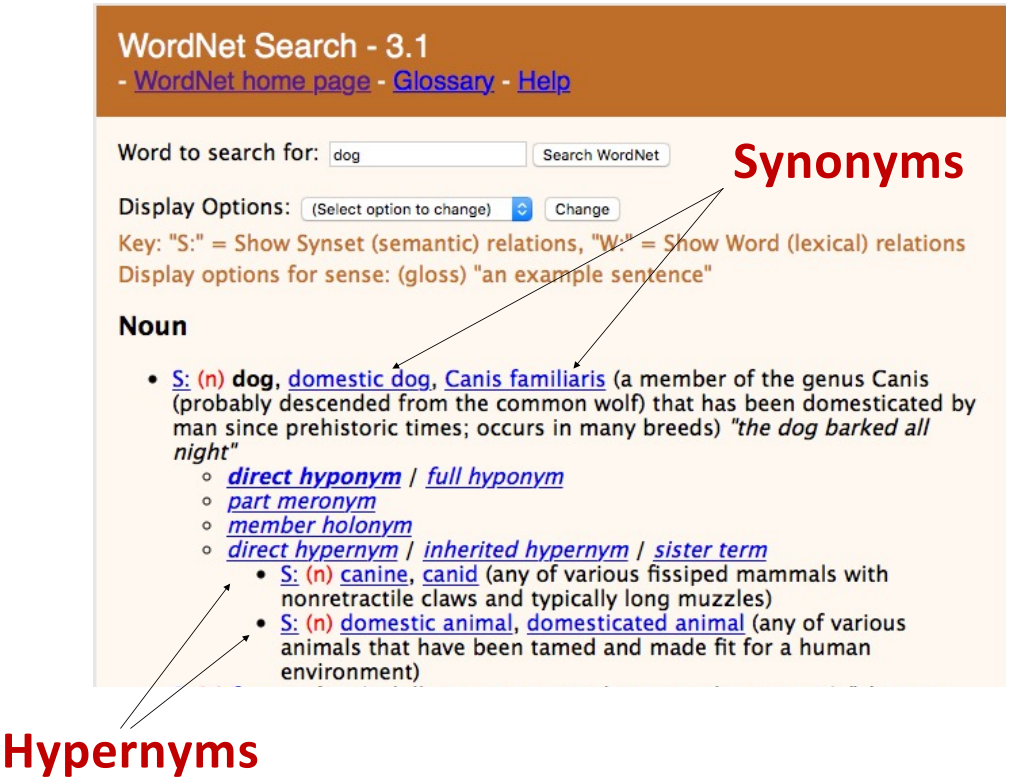
How can we represent word meaning?



- Traditional approach: symbolic lexical resources (e.g., WordNet)
- Represent meaning via explicit relations:
  - synonymy (dog ↔)
  - hypernymy (dog → animal)
- Meaning is **discrete**, **manually defined**, and **relational**

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Limitations of Symbolic Lexical Resources

- **Static and incomplete**
  - Cannot keep pace with new words, senses, or usage
- **Expensive to maintain**
  - Requires expert human annotation
- **Limited notion of similarity**
  - No natural notion of *graded* similarity
  - Cannot capture usage-based meaning differences

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Discrete Representation

- Words treated as atomic symbols (e.g., *one-hot encoding*).

motel [0, 0, 0, 0, 0, 1, 0, 0, 0, …, 0, 0]

hotel [0, 0, 1, 0, 0, 0, 0, 0, 0, …, 0, 0]

- **Vocabulary size = dimensionality** (can be very large)
- **Key limitation**: all words are equally dissimilar
  - cosine(motel, hotel) = 0

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Distributional Semantics

"You shall know a word by the company it keeps."

(J. R. Firth 1957)

*A **bank** is a financial institution that accepts deposits from the public and creates credit*

*In geography, the word **bank** generally refers to the land alongside a body of water.*

- Word meaning inferred from **contextual usage**
- Same word, different contexts → different meanings
- Meaning becomes statistical rather than symbolic

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Context-based Word Representations

- Build a word–context co-occurrence matrix
- Each word represented by its surrounding words
- **Rows** = target words; **Columns** = context words; **Values** = frequency counts

counts	I	like	enjoy	deep	learning	NLP	flying	.
I	0	2	1	0	0	0	0	0
like	2	0	0	1	0	1	0	0
enjoy	1	0	0	0	0	0	1	0
deep	0	1	0	0	1	0	0	0
learning	0	0	0	1	0	0	0	1
NLP	0	1	0	0	0	0	0	1
flying	0	0	1	0	0	0	0	1
.	0	0	0	0	1	1	1	0

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Issues with raw co-occurrence count

- Extremely high-dimensional
- Sparse and memory-intensive
- Sensitive to corpus size and frequency effects

counts	I	like	enjoy	deep	learning	NLP	flying	.
I	0	2	1	0	0	0	0	0
like	2	0	0	1	0	1	0	0
enjoy	1	0	0	0	0	0	1	0
deep	0	1	0	0	1	0	0	0
learning	0	0	0	1	0	0	0	1
NLP	0	1	0	0	0	0	0	1
flying	0	0	1	0	0	0	0	1
.	0	0	0	0	1	1	1	0

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Low Dimensional Dense Word Embeddings

- Represent each word as a **dense vector**
- Fixed dimensionality (typically **100–300 for word2vec**)
- Learned **automatically from data**

Core Idea of word2vec

- Learn word embeddings by **predicting context**
- Replace explicit co-occurrence counts with a **prediction task**

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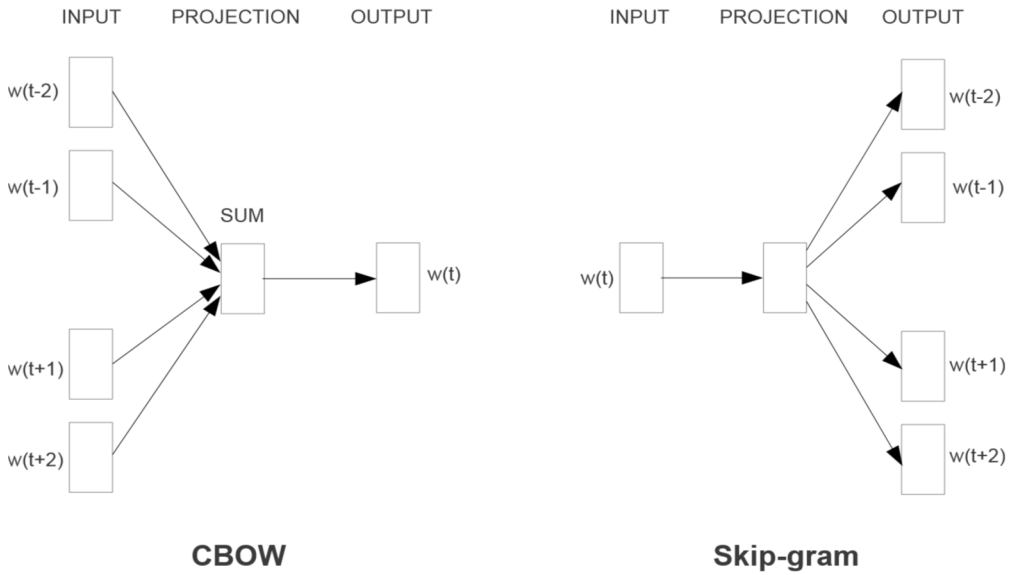
Word2vec Architectures

- Two training objectives
  - **CBOW**: predict target word from context
  - **Skip-gram**: predict context from target word



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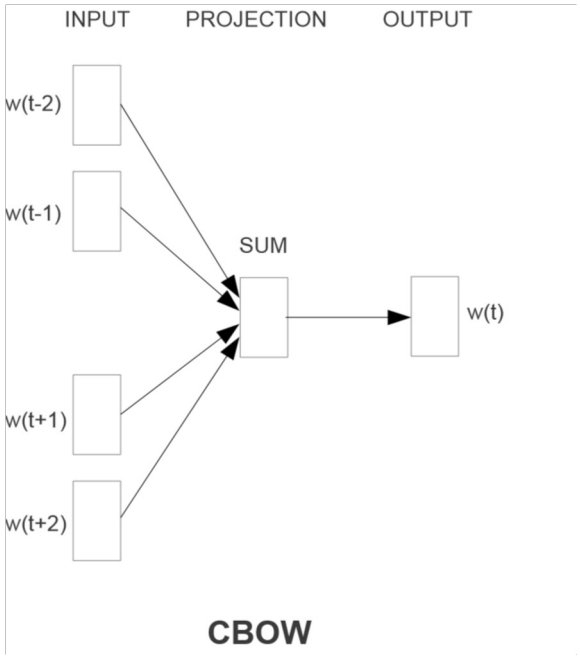
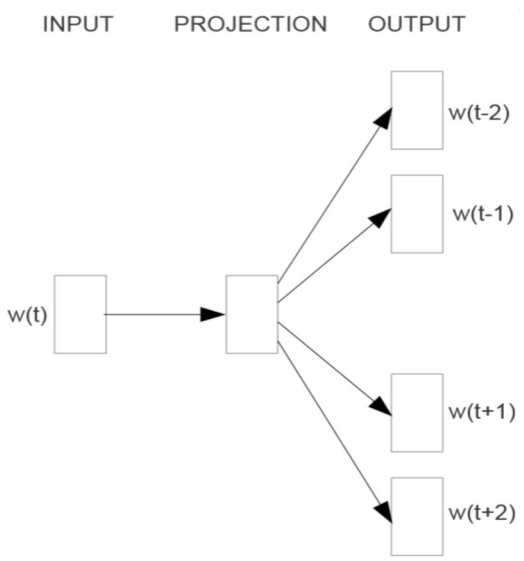
Continuous Bag-of-Words (CBOW)



- **Input**: surrounding context words
- **Output**: target word
- Context vectors are **summed or averaged**
- Efficient for frequent words

"Given the context, what word fits here?"

Skip-gram



Skip-gram

- **Input**: target word
- **Output**: surrounding context words
- Generates multiple training pairs per word
- Often performs better than CBOW for rare words

"Given this word, what words tend to appear nearby?"

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CBOW vs. Skip-gram

CBOW: predict *apple* from {eat, an, every, day}

Skip-gram: predict {eat, an, every, day} from *apple*

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Training objective (CBOW)

- Objective function

$$\prod_{w \in C} P(w \mid \text{context}(w))$$

- Taking logarithm, we have

$$\mathcal{L} = \sum_{w \in C} \log P(w \mid \text{context}(w))$$

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Applications of Word Embeddings

- Dependency parsing
- Named entity recognition
- Document classification
- Sentiment Analysis
- Paraphrase Detection
- Word Clustering
- Machine Translation

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Word Analogies

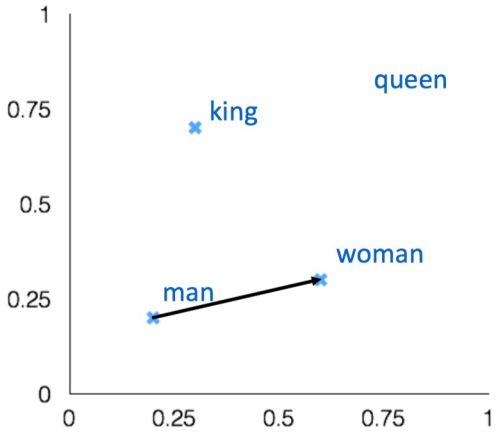
- Test for **linear regularities** in embedding space
- Relations captured as vector offsets
- Empirical property observed by Mikolov et al. (2014)

a:b :: c:?

$$d = \arg\max_x \frac{(w_b - w_a + w_c)^T w_x}{||w_b - w_a + w_c||}$$

man:woman :: king:?

+	king	[0.30 0.70]
-	man	[0.20 0.20]
+	woman	[0.60 0.30]
	queen	[0.70 0.80]



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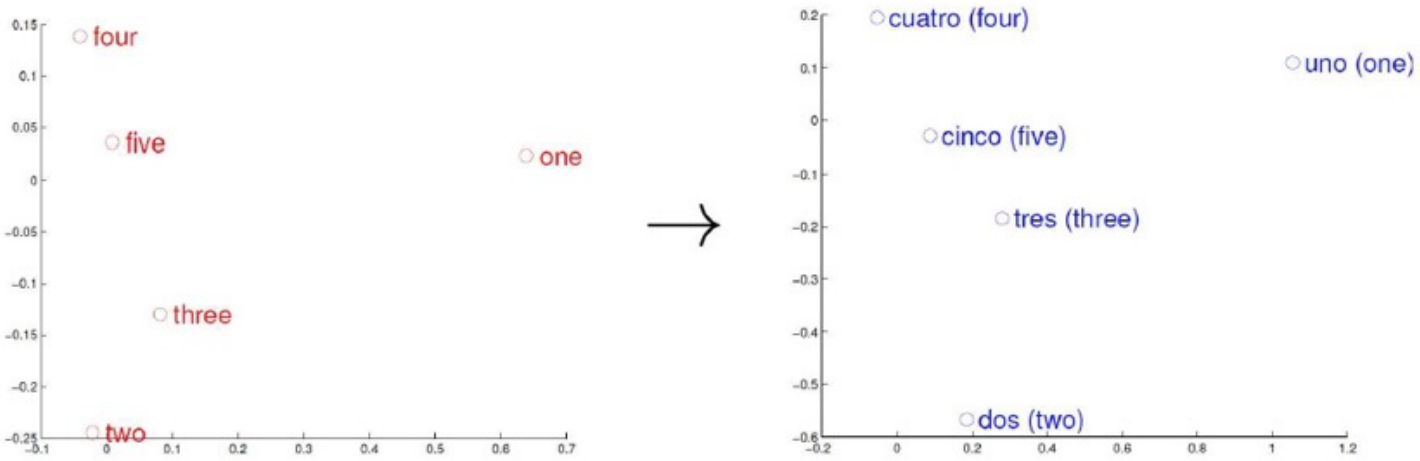
Machine Translation

- Similar concepts align geometrically across languages
- Enables bilingual lexicon induction
- Precursor to modern multilingual embeddings



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Evolution of Language Models

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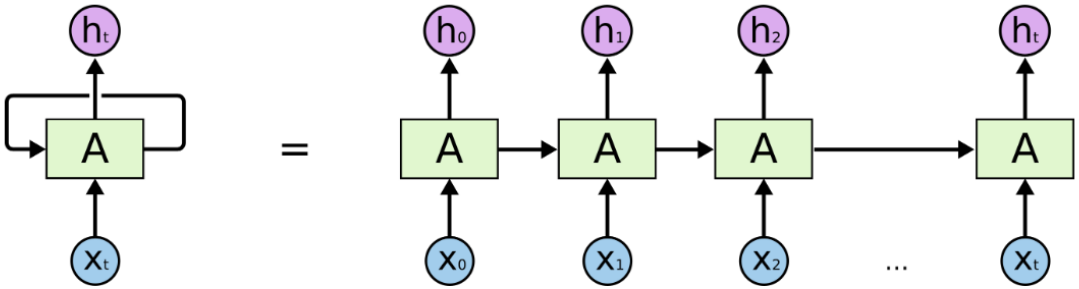
Recurrent Neural Networks (RNNs)



An unrolled recurrent neural network.

Source: http://clah.github.io/posts/2015-08-Understanding-LSTMs/

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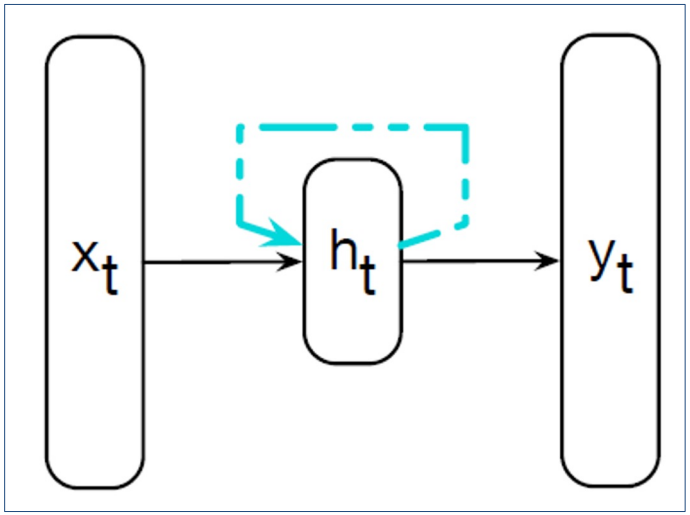
Recurrent Neural Networks (RNNs) as LMs

Recurrent Neural Network

- A network that contains a cycle, i.e., the hidden state is fed back and used at the next time step
- The hidden layer has a recurrent connection: the hidden state at time t depends on
  – the current input x_t, and
  – the hidden state from the previous time step h_{t-1}



An Elman network (simple RNN)

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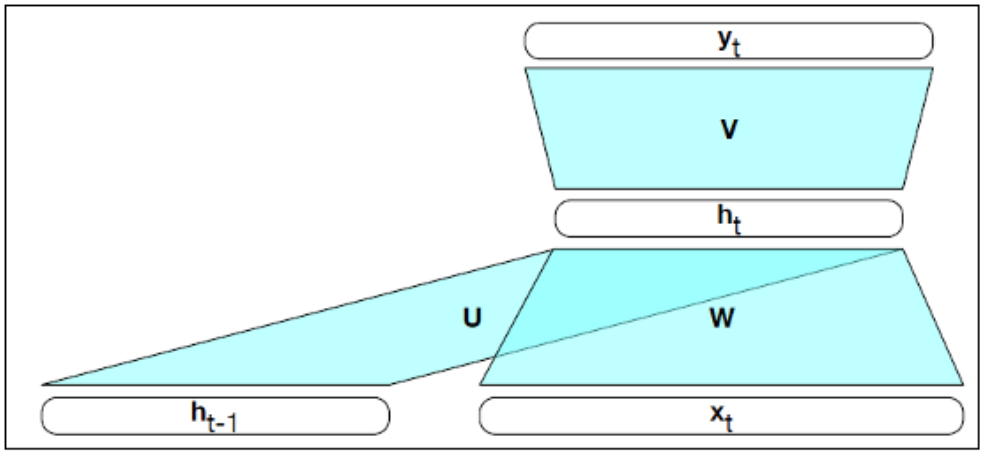
Recurrent Neural Networks (RNNs) as LMs

Simple RNN

- The sequence is processed one token at a time, from left to right
- At each time step t, the hidden state is updated using:
  – the current input x^t
  – the previous hidden state $h_{(t-1)}$
- The hidden state h_t serves as a summary of past inputs and is used to compute the output at time t.



A closer look at a simple RNN

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Recurrent Neural Networks (RNNs) as LMs

- RNN language models
  - Process an input sequence one word at a time
  - To predict the next word from the current word and the previous hidden state
  - Model the next-word probability
- Computation is sequential, limiting parallelization

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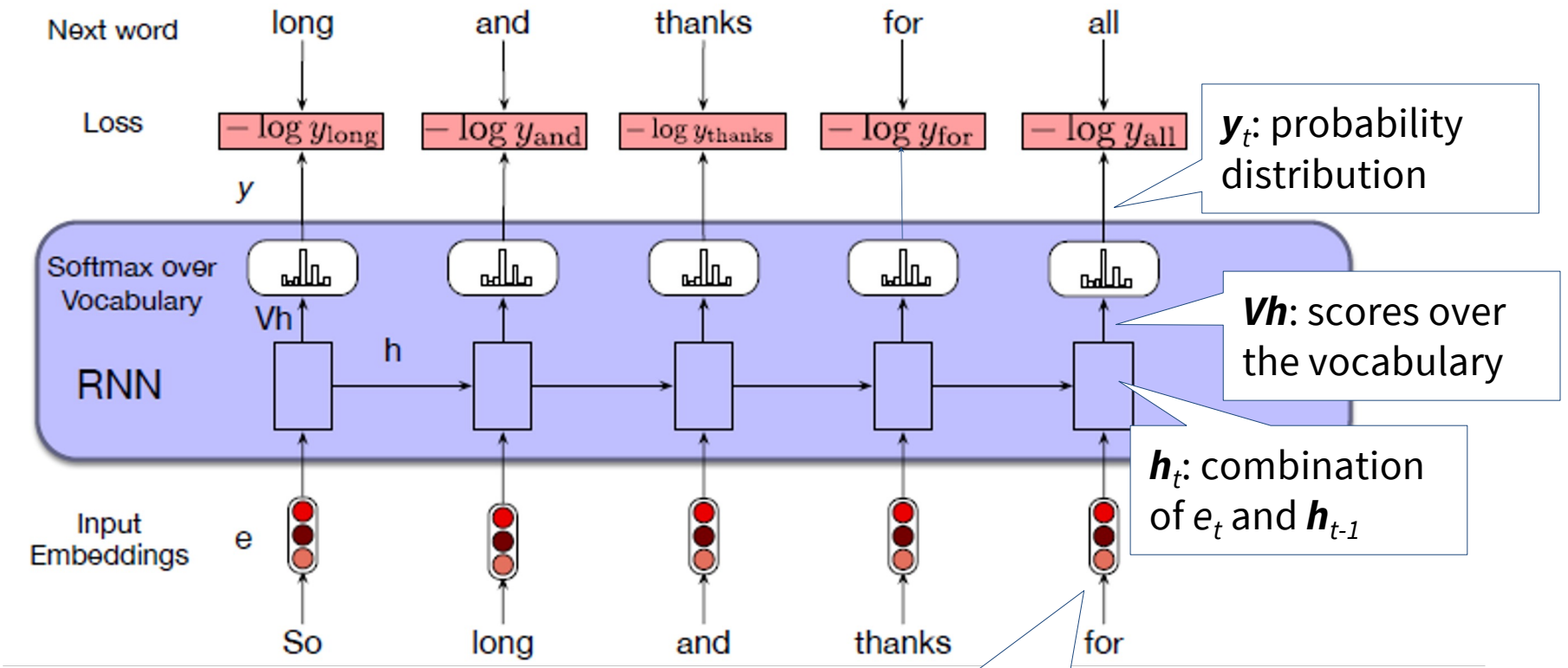
Recurrent Neural Networks (RNNs) as LMs



Let *t* be the current time step; w_t is the current word

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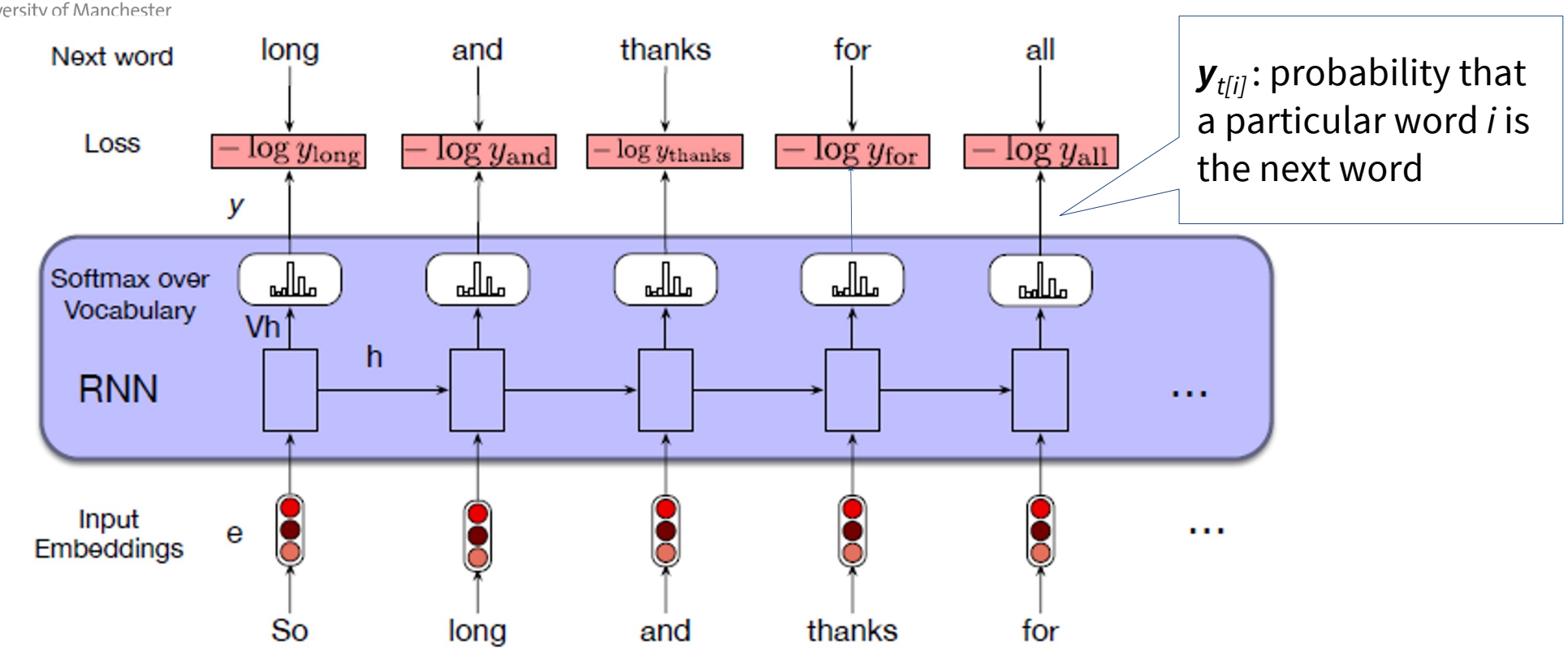
Recurrent Neural Networks (RNNs) as LMs



Let *t* be the current time step; w_t is the current word

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Recurrent Neural Networks (RNNs) as LMs

- Training an RNN as a language model
  - based on a corpus, model learns to predict the next word at each time step t
  - model minimises the error in predicting the correct next word
  - error/loss: the difference between a predicted probability distribution and the correct distribution, i.e., cross-entropy, where:

correct distribution: one-hot vector for the vocabulary, where the actual next word = 1 and the rest are 0

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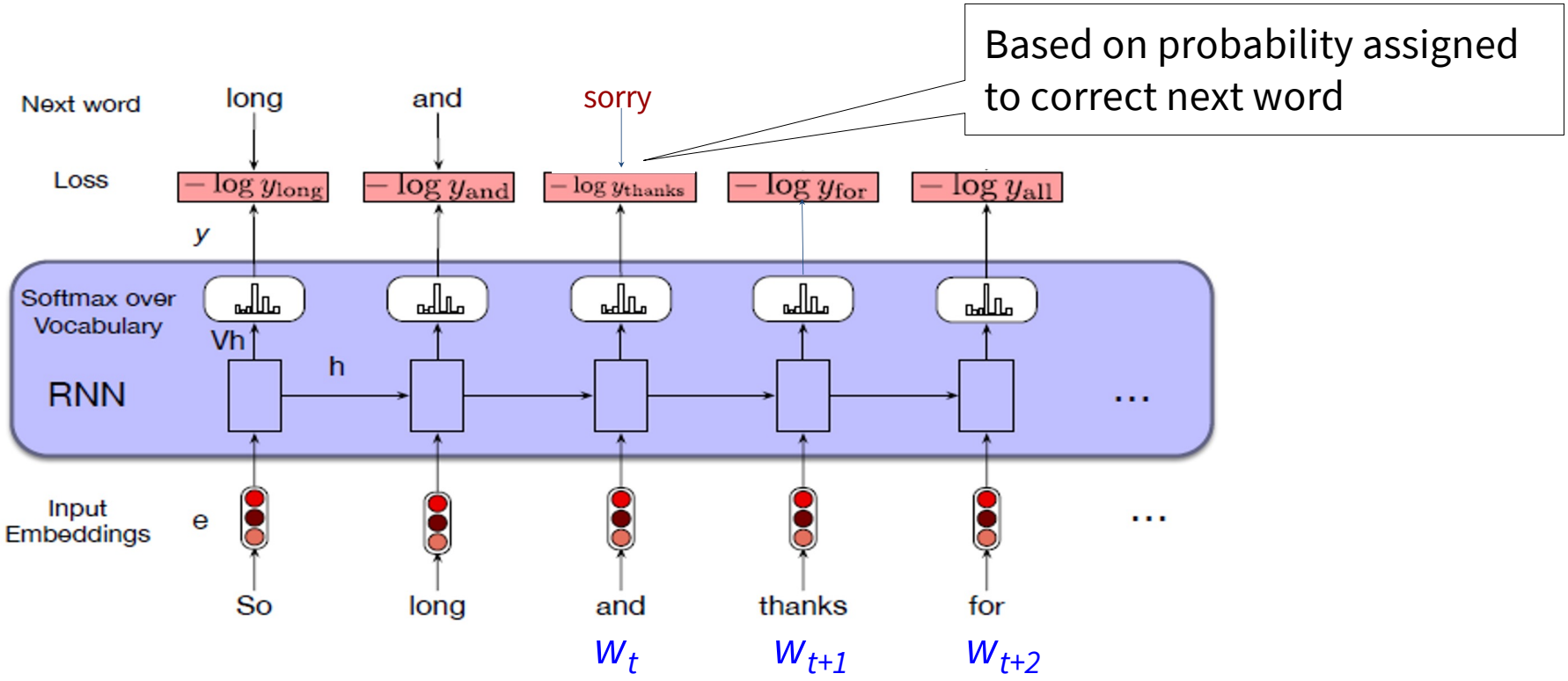
Recurrent Neural Networks (RNNs) as LMs

Teacher forcing: the model is given the correct history sequence (rather than what was predicted in the previous time step)



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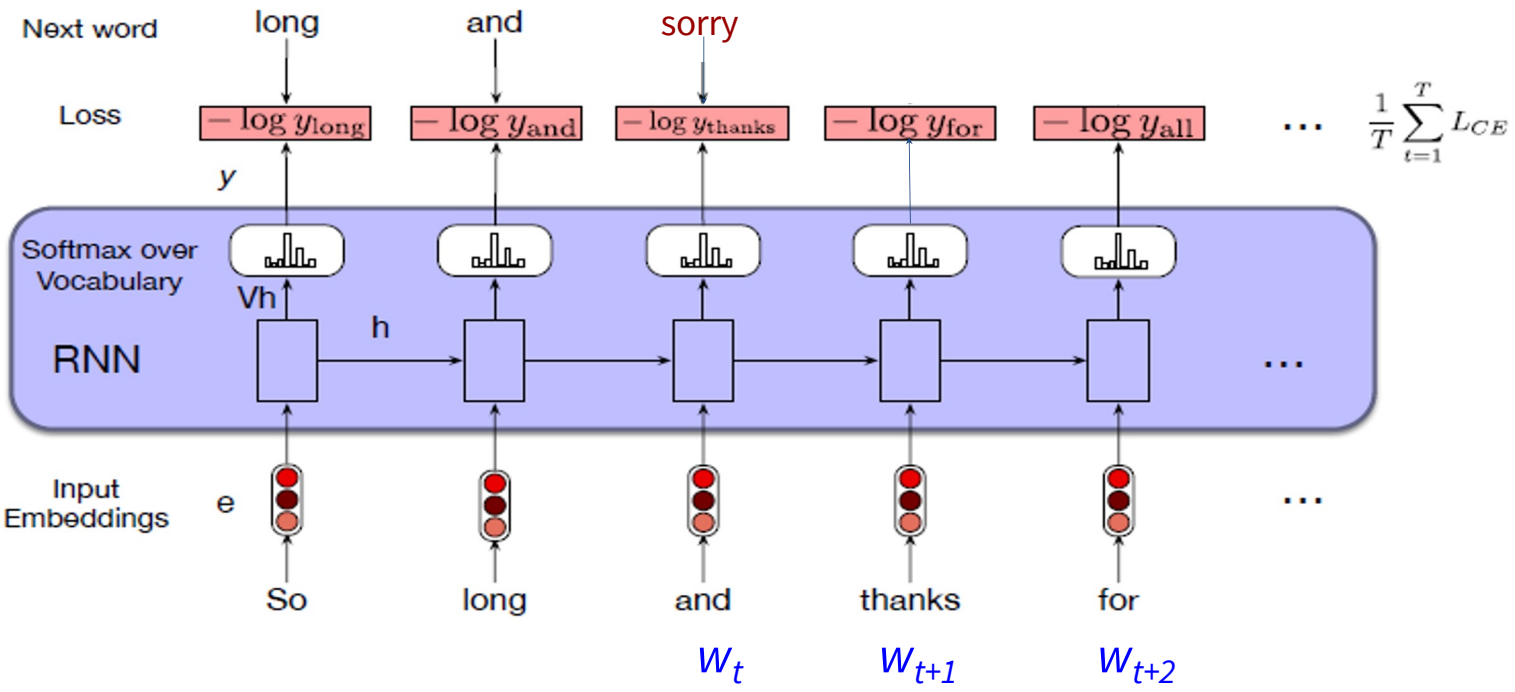
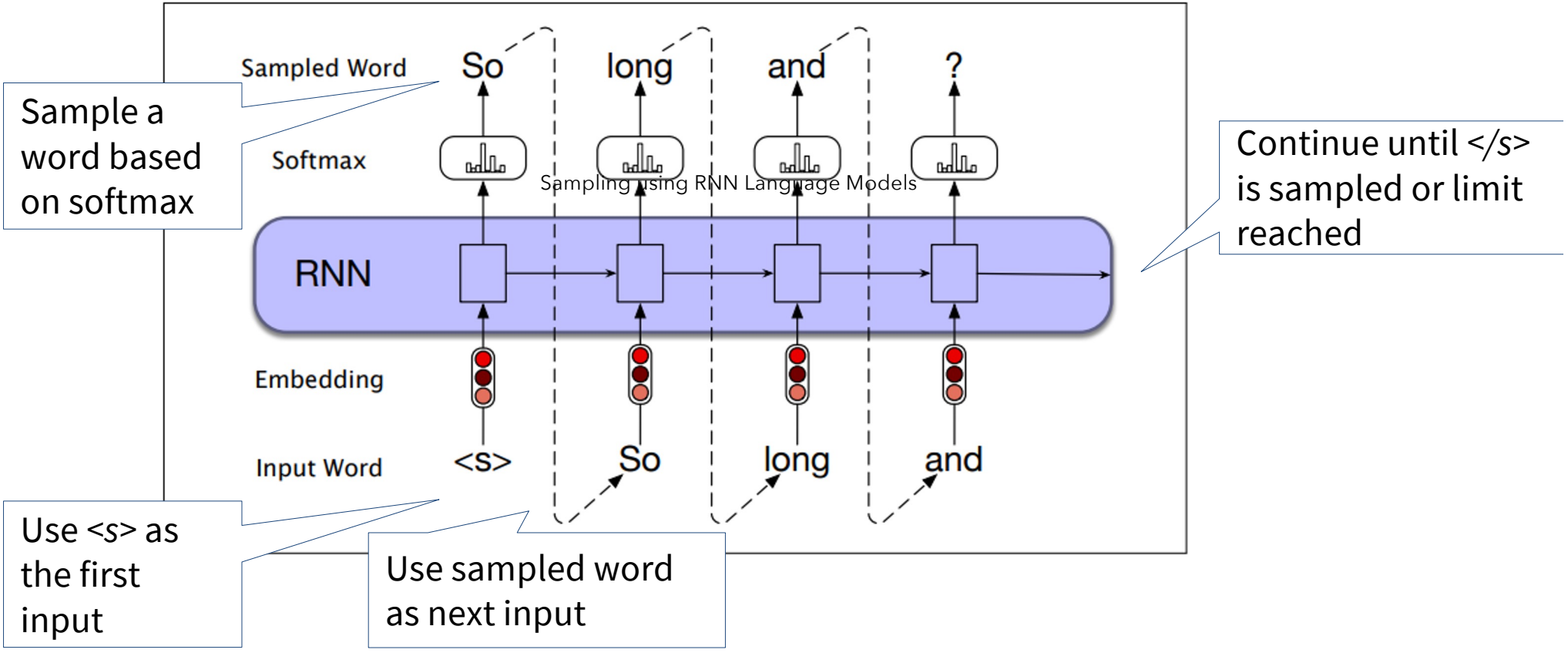
Recurrent Neural Networks (RNNs) as LMs

Gradient descent: used to adjust the weights in the RNN, in order to minimise the CE loss (L_{CE}) averaged over the sequence



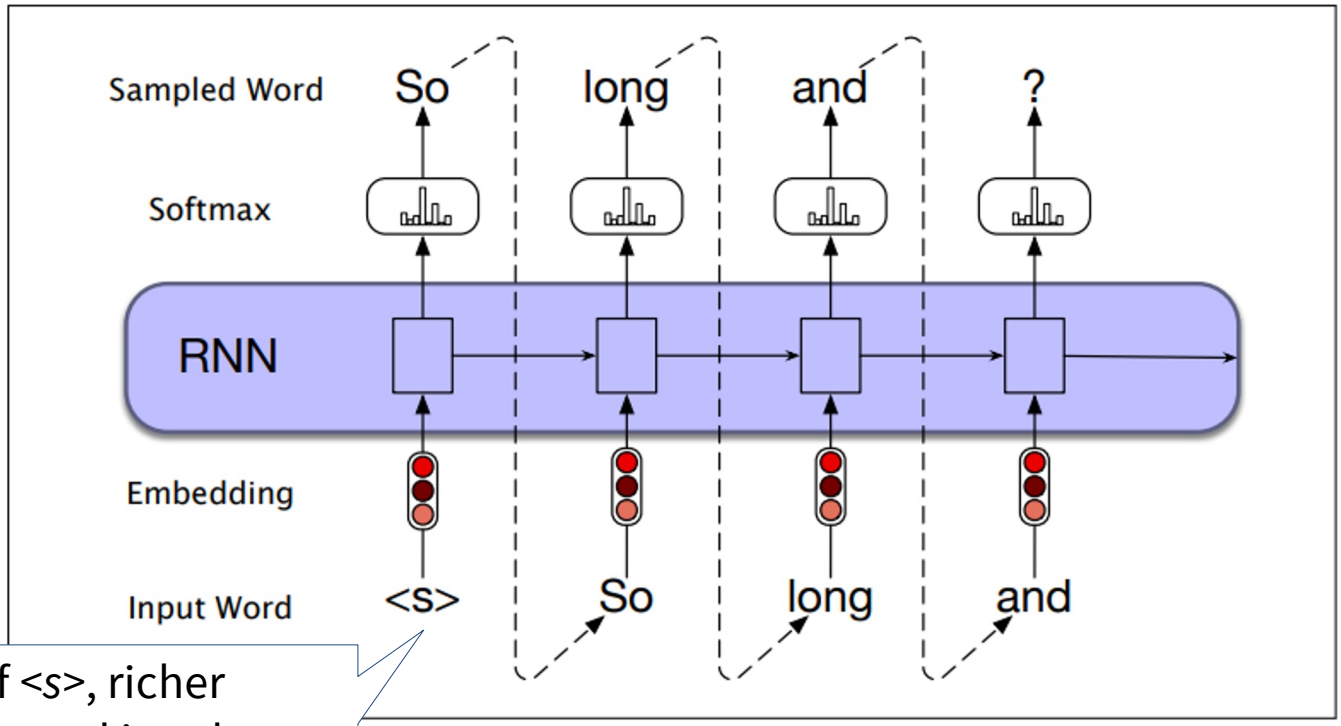
Sampling using RNN Language Models

Autoregressive generation: the word generated at each time step t is conditioned on the word selected by the RNN from the previous time step t-1



Sampling using RNN Language Models

Autoregressive generation: used in many other applications (apart from predicting the next word), e.g., machine translation, summarisation



Instead of <s>, richer context is used in other applications

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Quick Comparison

- Key difference between *n*-gram and RNN language models:
  - *n*-gram language models incorporate information from limited context only (i.e., preceding n-1 tokens)
  - RNN language models: the hidden state can incorporate information from all the preceding words

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Limitations of RNNs

- Process input sequentially, slow training and inference
- Struggle with long-range dependencies despite LSTM/GRU improvements.
- Limited parallelisation during training, so hard to scale on large datasets.
- Gradient vanishing/exploding issues over long sequences.
- Difficult to capture global context effectively

There is another type, i.e., transformer language models, up next!

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Turning Point – Transformer

- Proposed by Vaswani et al. (2017) in "*Attention is All You Need*".
- Eliminates recurrence, i.e., no step-by-step token processing.
- Enables parallelization: trains faster by processing entire sequences simultaneously.
- Leverages *self-attention* to model long-range dependencies.
- Achieves SOTA results in machine translation and beyond

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Evolution of Language Models

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Transformer

Encoder Only

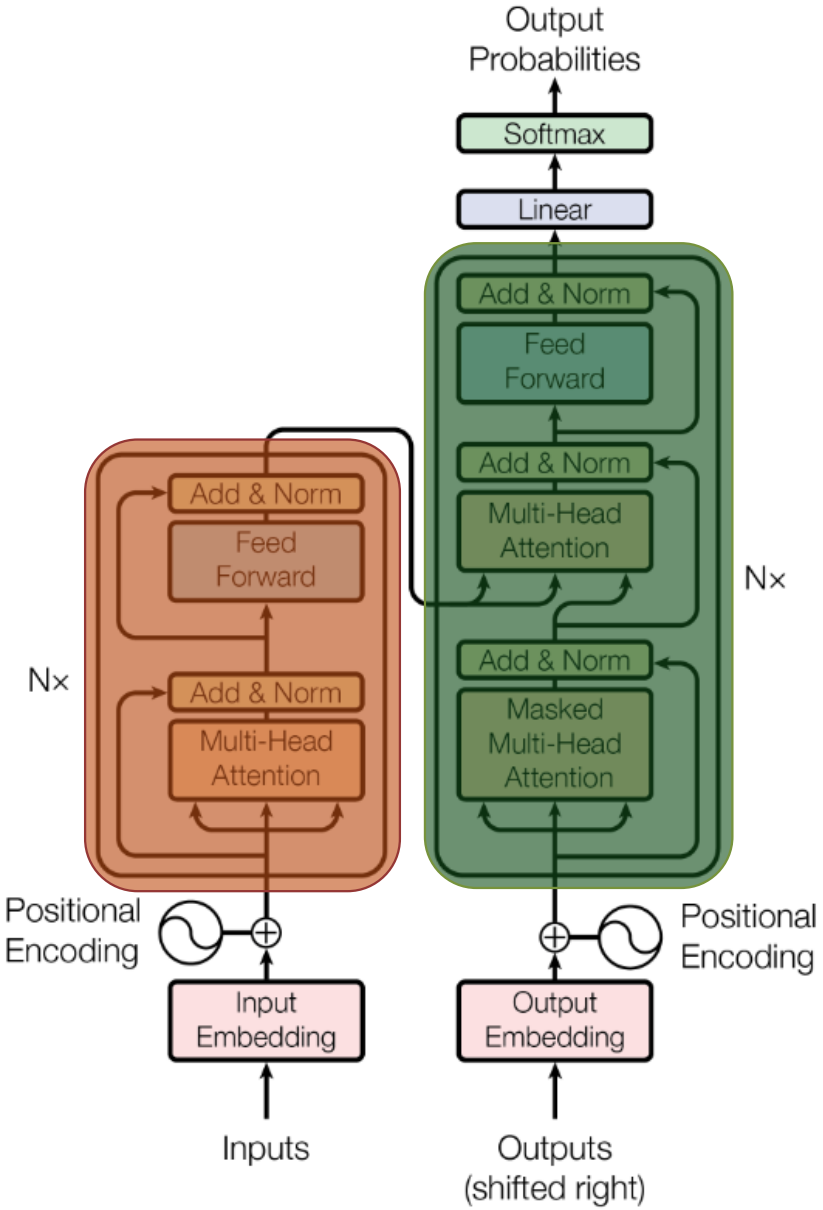
Uses: NLU, classification, feature extraction
Examples: BERT, RoBERTa, DeBERTa

Encoder-Decoder

Uses: NLG, translation, summarisation, seq2seq mapping
Examples: T5, Flan-T5, BART

Decoder Only

Uses: NLG, translation, summarisation, completion
Examples: GPT-x, LLaMa, every other new model today



Transformers

Transformer: a neural architecture built entirely on attention mechanisms.

Credits: The Illustrated Transformer – Jay Alammar

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Transformer Architecture

Core Components

- **Encoder Stack**: produces contextual representations of the input.
- **Decoder Stack**: uses the encoder's output and previous decoder outputs to generate the target sequence.
- **Self-Attention Mechanism**: allows the model to focus on different parts of the input sequence when processing each token.
- **Feed-Forward Neural Networks**: apply nonlinear transformation per token.

***Key idea**: no recurrence, no convolution — only attention.*

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Why Self-Attention Replaced RNNs

- In RNNs:
  - Information between distant tokens flows through $O(n)$ steps
  - Computation is sequential, i.e. hard to parallelize
- In Transformers:
  - Self-attention allows each token to directly attend to all other tokens in the sequence, producing contextualised representations
  - Computation within each layer is parallelizable across sequence positions
- Trade-off:
  - Attention time and memory scale **quadratically** with sequence length $O(n^2)$

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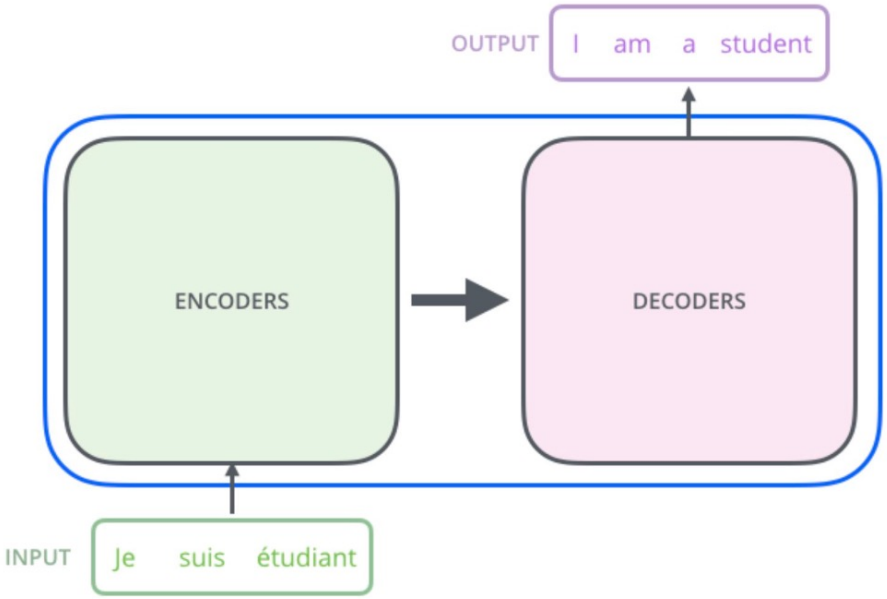
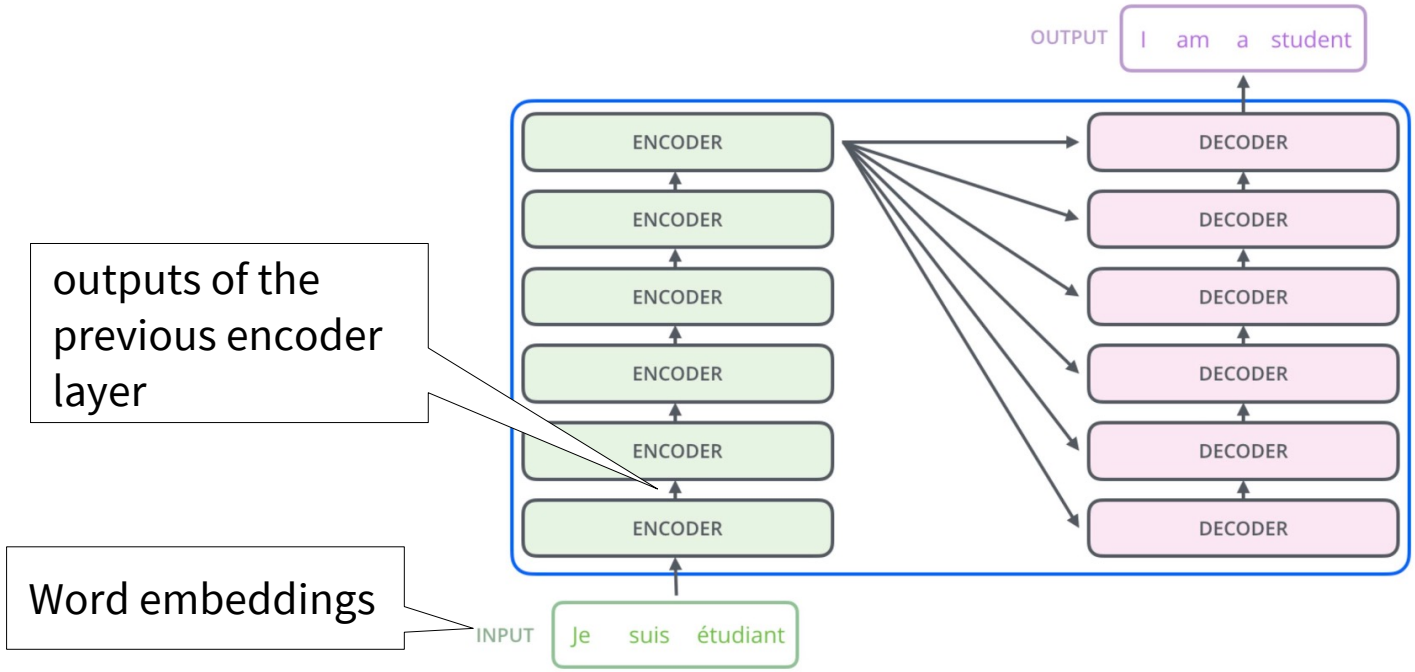
Transformer Architecture

- Opening the box with more details …



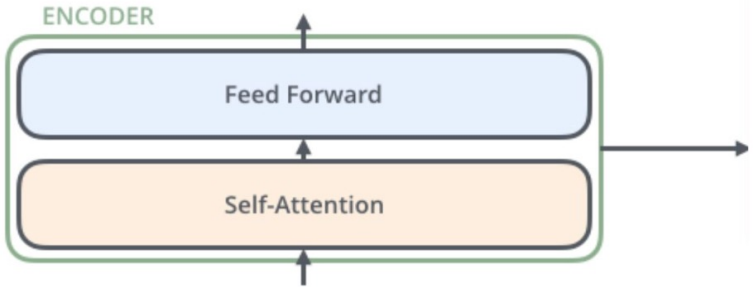
Transformer Architecture

- Opening the box with even more details …

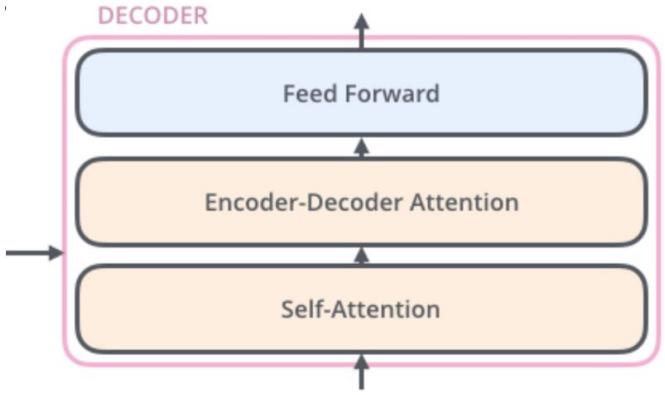


Transformer Architecture

Encoder:



Decoder:



Encoder:

- Self-attention + Feedforward
- Produces contextual representations of input

Decoder:

- Masked self-attention
- Cross-attention (attend to encoder outputs)
- Generates output one token at a time

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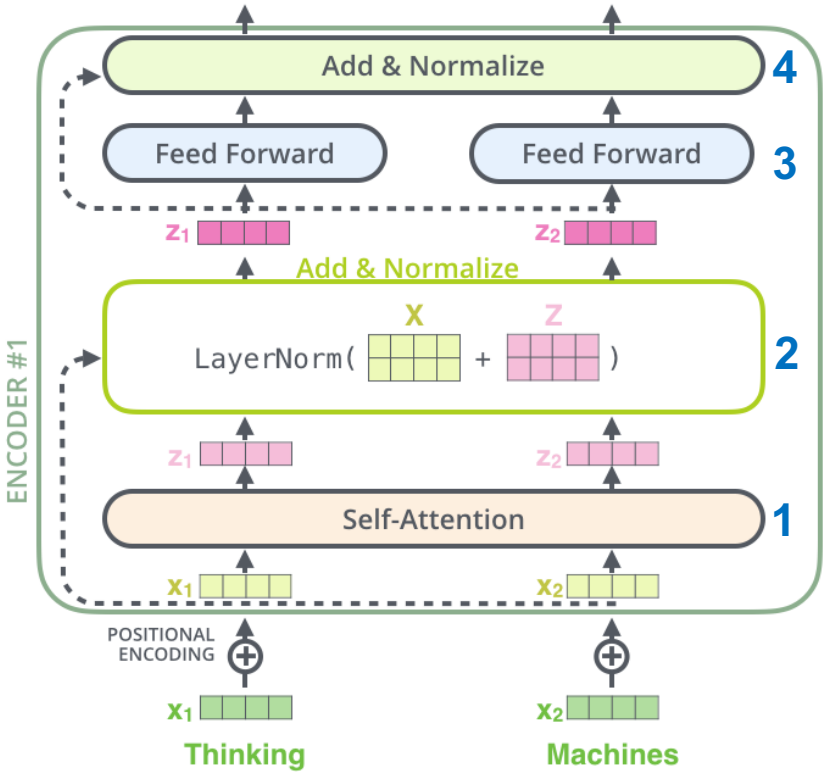
Inner workings of the Encoder



1. **Self-attention** mixes information across positions.
2. **Residual** + **LayerNorm**
3. **FNN** (position-wise) applies nonlinear transformation independently to each token.
4. **Residual** + **LayerNorm**

$$\text{LayerNorm}(x + \text{SelfAttention}(x))$$

$$\text{LayerNorm}(z + \text{FFN}(z))$$

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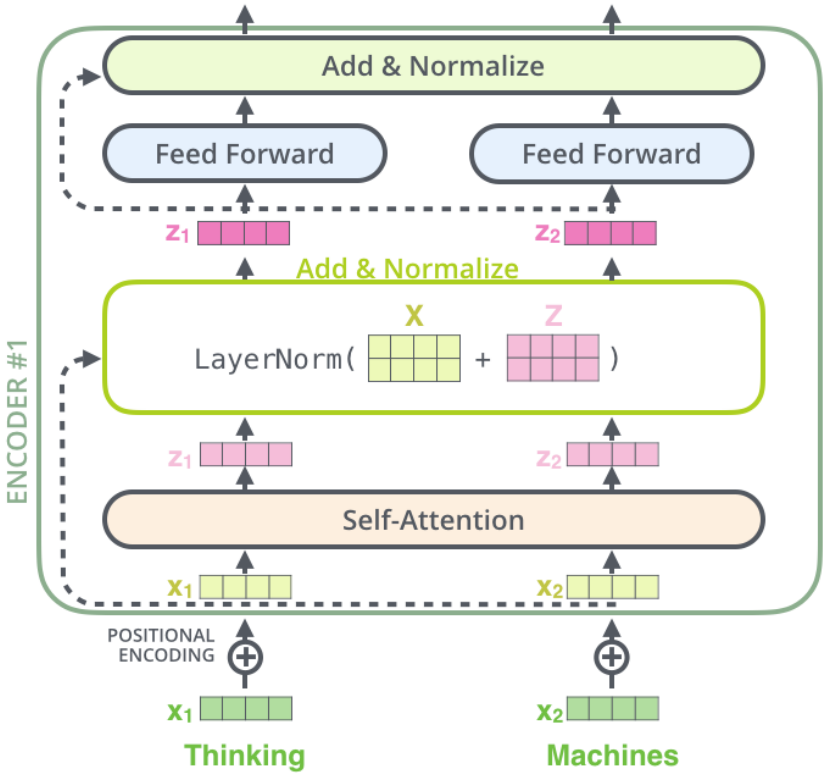
Inner workings of the Encoder



Important properties

- Tokens are processed in parallel within each layer
- Tokens are coupled via attention
- Deeper layers refine contextual representations
- Residual connections preserve signal and ease optimisation.

$$\text{LayerNorm}(x + \text{SelfAttention}(x))$$

$$\text{LayerNorm}(z + \text{FFN}(z))$$

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Self-Attention Mechanism

- Scaled Dot-Product Attention
- For input $X \in \mathbb{R}^{n \times d}$:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- Interpretation:
  - *Q*, *K*, *V* are learned linear projections of the input
  - Attention scores $\frac{QK^T}{\sqrt{d_k}}$ measure similarity in projected space
  - Scaling $\sqrt{d_k}$ prevents dot products from growing large in high dimensions and stabilises gradients.
  - softmax converts attention scores into a distribution over the input elements
  - Output is a weighted sum of value vectors

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Understanding Self-Attention

- **Query (Q)**: The query vector represents the current input or the element for which we are trying to find relevant information.

  Example: Looking for action movies with strong female leads.

- **Key (K)**: The key vector represents the criteria or features against which the query is compared. Each element in the input data has an associated key vector.

  Example: This movie is tagged as action, drama, female lead.

- **Value (V)**: The value vector holds the actual information or content that might be relevant to the query.

  Example: The full movie details or recommendation result.

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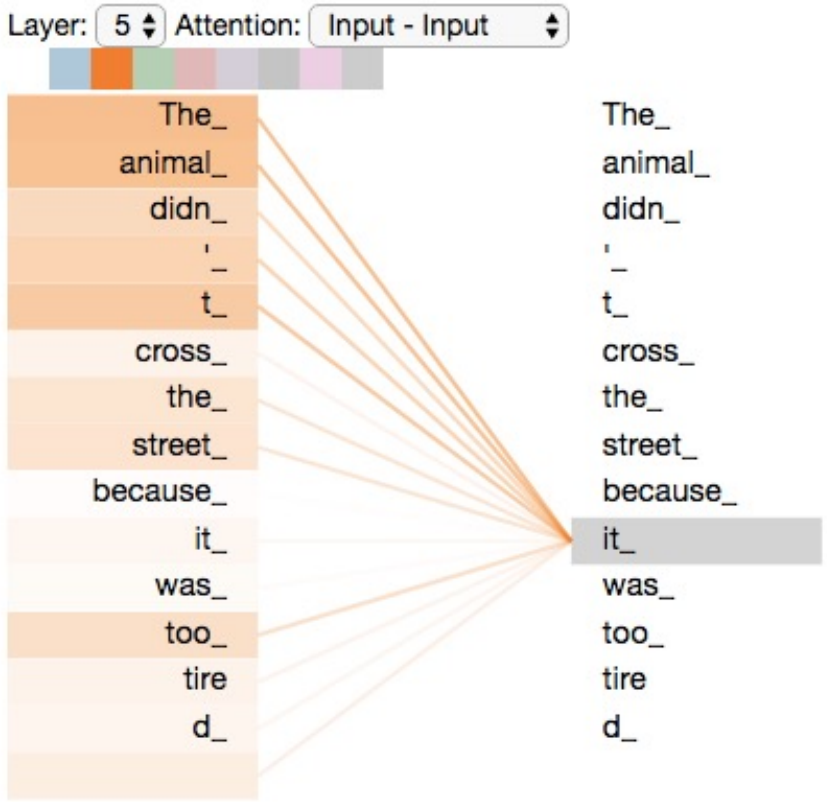
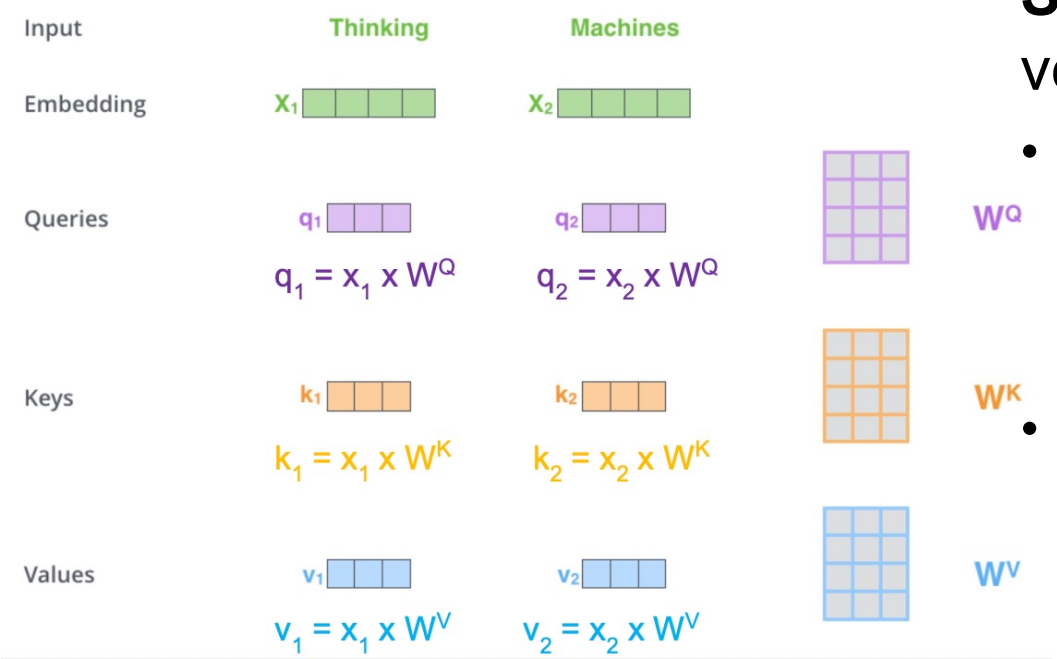
Self-attention Visualisation



- Self-attention allows each token focus on relevant token in the sequence.
- Token "*it*" attends more to related tokens like "*The animal*"
- This helps the model capture long-range dependencies and context.

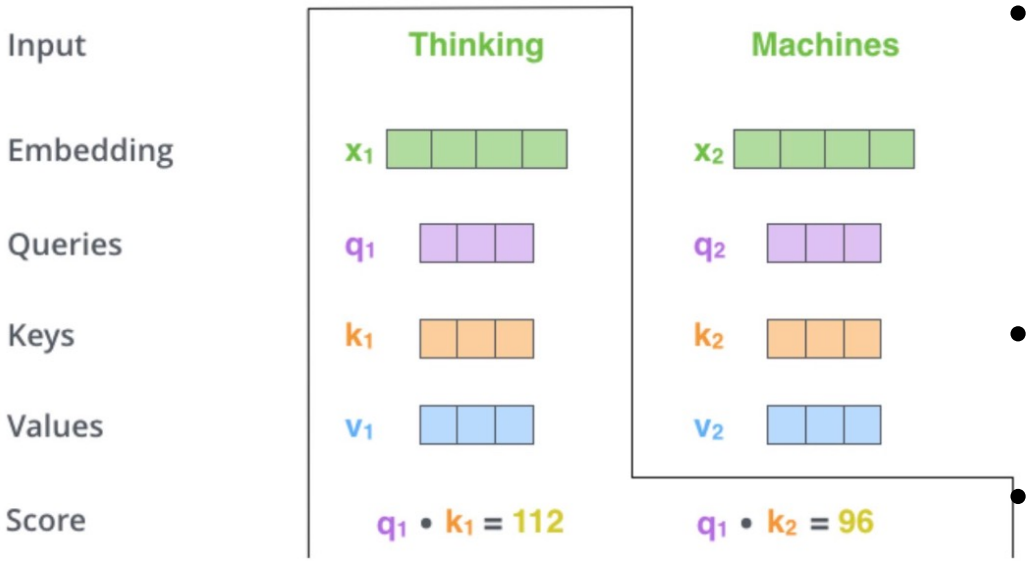
Calculating Self Attention



Step 1: Compute ***Q***, ***K***, and ***V*** vectors

- For each token, compute
  - a **query** vector
  - a **key** vector
  - a **value** vector.
- Derived by multiplying input embeddings with learned weight matrices

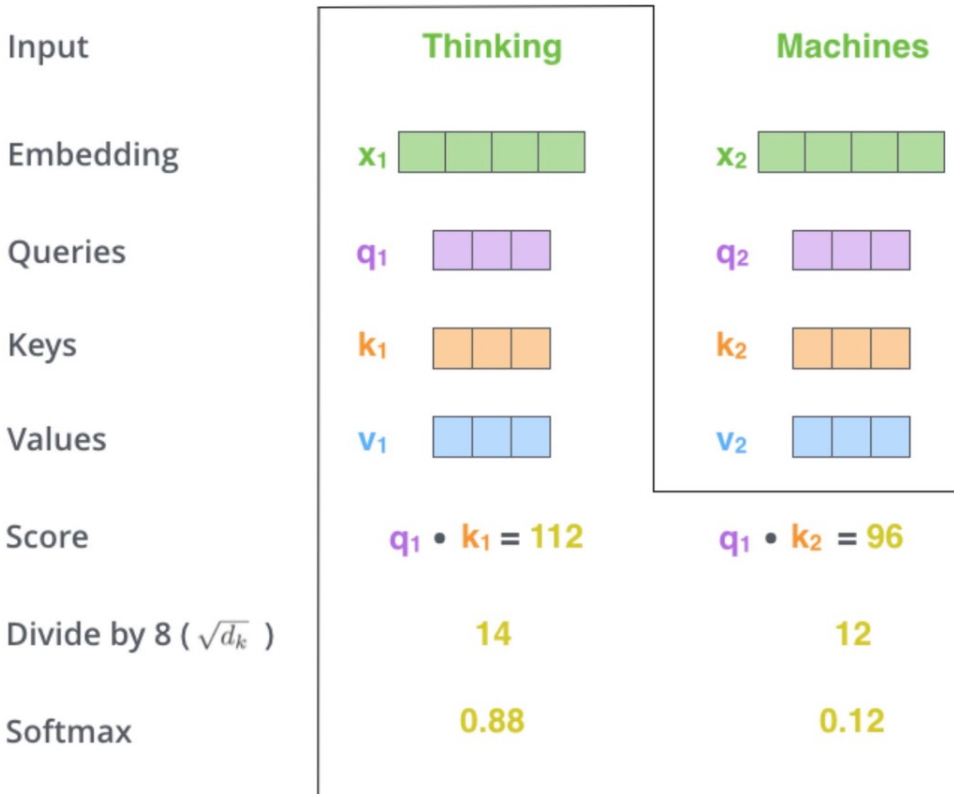
Calculating Self Attention



Step 2: Compute attention scores

- For each token, compute dot products between
  - Its query vector and all key vectors
  - e.g. $\boldsymbol{q_1}$ with $\boldsymbol{k_1}$, $\boldsymbol{k_2}$, $\boldsymbol{k_3}$, …
- This produces one score per token pair
- Larger dot product → stronger alignment

Calculating Self Attention



Step 3: Scale and normalize attention scores

- The dot product scores from Step 2 can become large, especially with high-dimensional vectors.
- Divide each score by the square root of the key dimension (e.g. $\sqrt{64} = 8$) to prevent large magnitudes
- This scaling helps stabilize gradients during training.
- Apply Softmax
  - All weights are positive
  - Converts to a probability distribution over tokens

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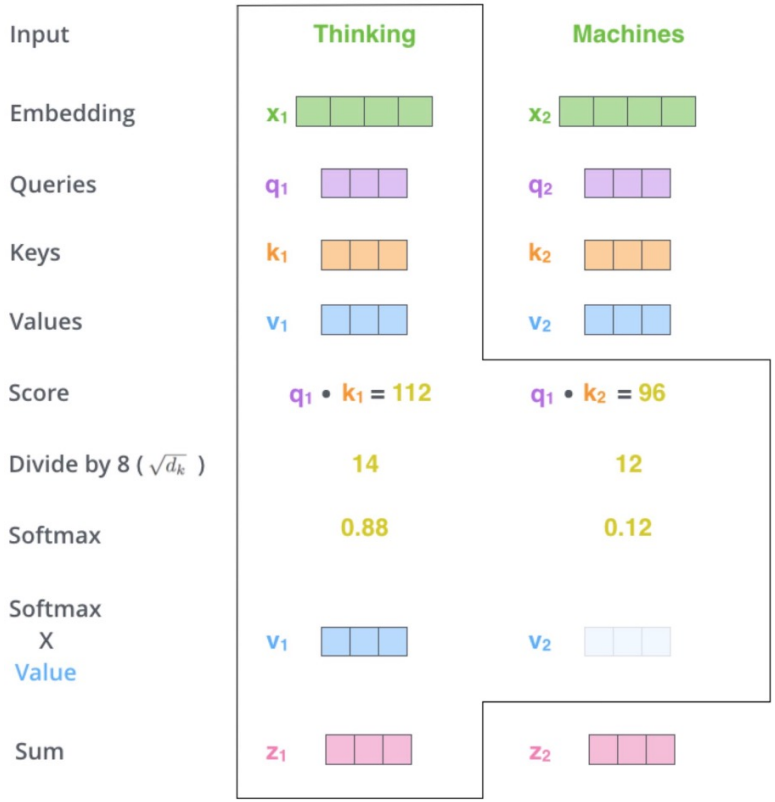
Calculating Self Attention

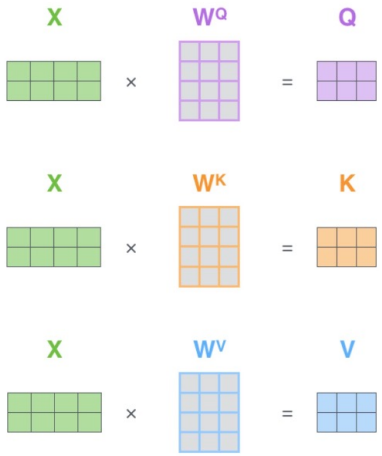


Step 4: Compute the self-attention output

- Multiply each value vector by its attention weight.
- This *emphasizes important* words and *down-weights irrelevant* ones.
- The result is the updated contextualized representation for the token
- This representation is then passed to the feed-forward layer for further processing

Efficient computation in matrix terms

- Calculate the *Query*, *Key*, and *Value* matrices for the entire context. We do that by packing the embeddings into a matrix X and multiplying it by the weight matrices.



- Outputs of the self-attention layer:

$$Z = \text{softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) V$$

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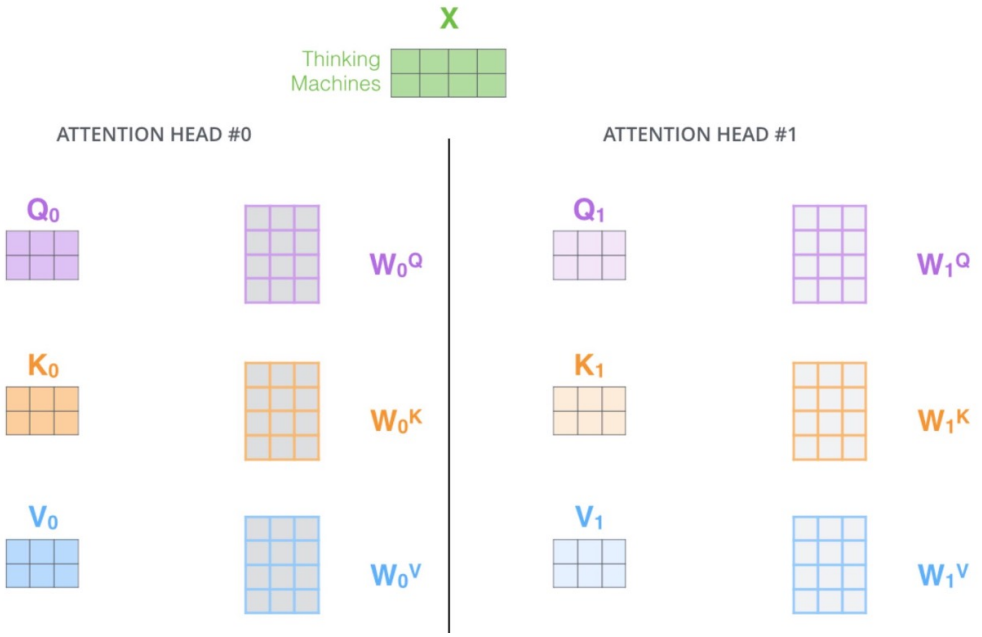
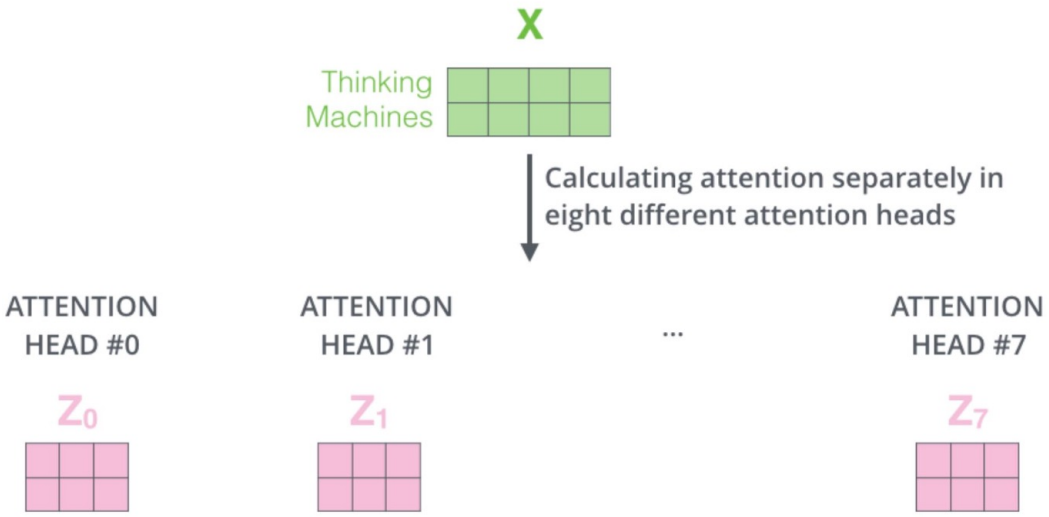
A few more ornaments: Multiple heads!



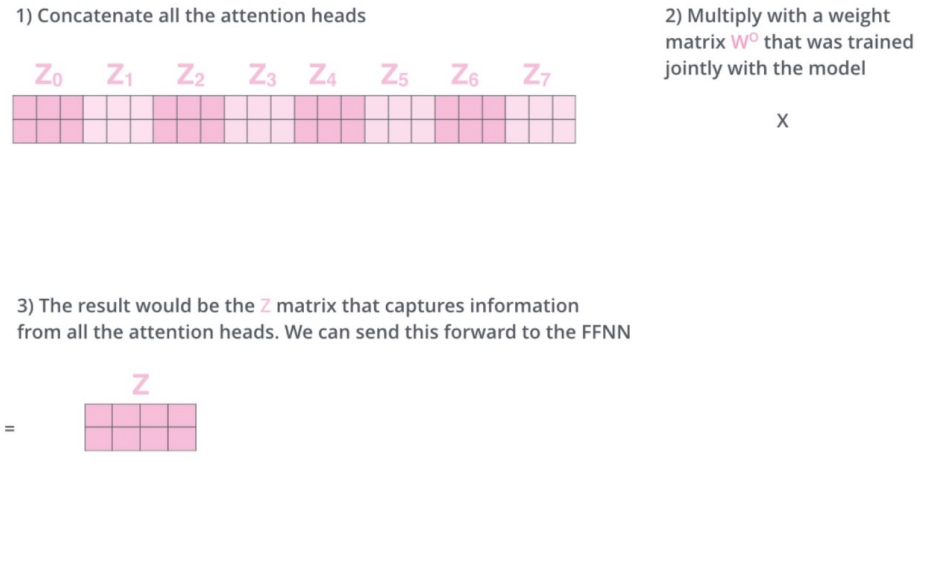
- Each self-attention layer computes multiple attention heads in parallel.
- Instead of one set of Q/K/V weights, we use multiple sets one per head, i.e. "***multi-head attention***"
- Each head learns to capture different aspects or patterns in the data.
- **Intuition**: enables the model to attend to different representation subspaces simultaneously (e.g. *river 'bank'* vs. *financial 'bank'*)

A few caveats: Multiple heads!



- Each attention head produces its own output matrix (Z_0, Z_1, ..., Z_7).
- These outputs are computed independently in different learned projection subspaces and can model diverse relational patterns from the same input.

Combining Outputs from Multiple Heads



- Concatenate the outputs (Z_0, Z_1, ..., Z_7) from all attention heads.
- Multiply the combined matrix by a learnable output projection matrix *Wo*.
- This gives the final output *Z*, which integrates diverse attention patterns and is passed to the next layer.

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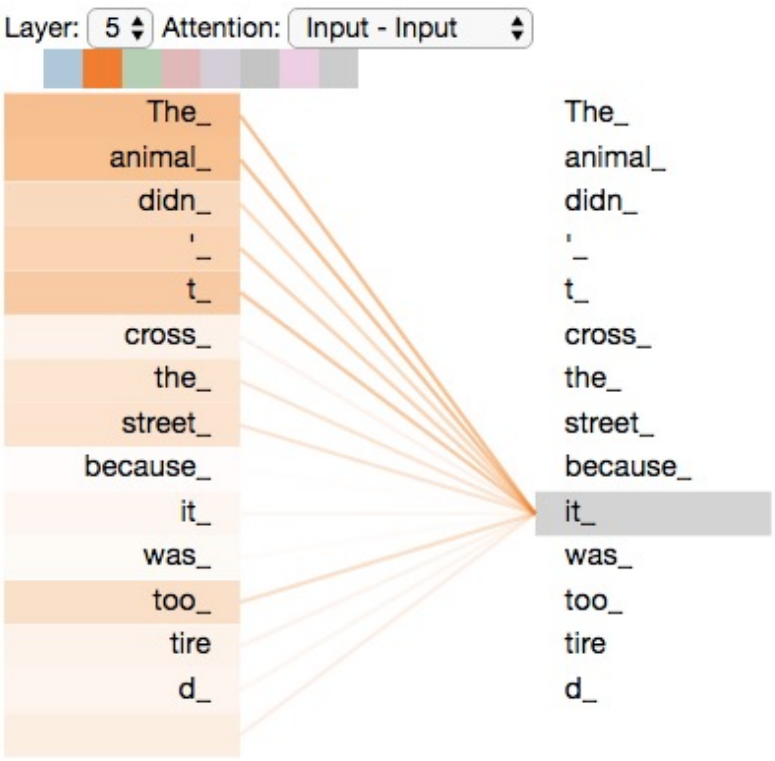
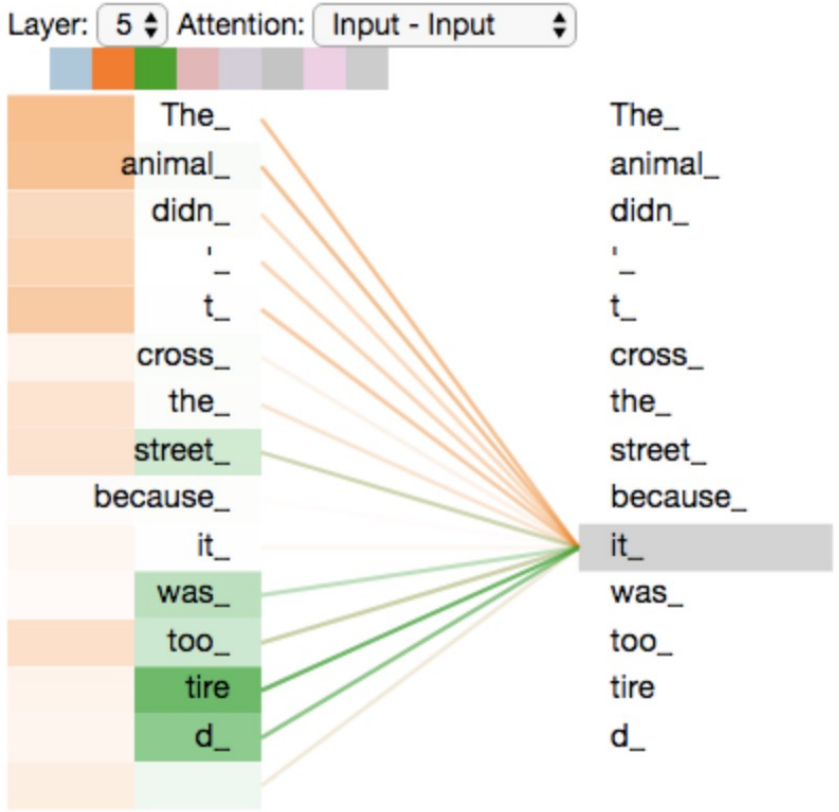
Multi-Headed Attention





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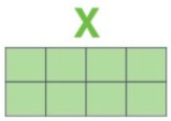
All in one place…

1) This is our input sentence*

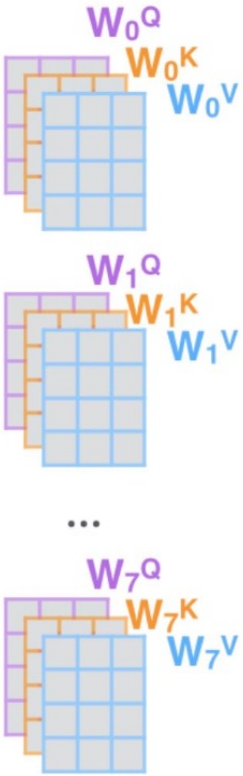
Thinking
Machines

2) We embed each word*

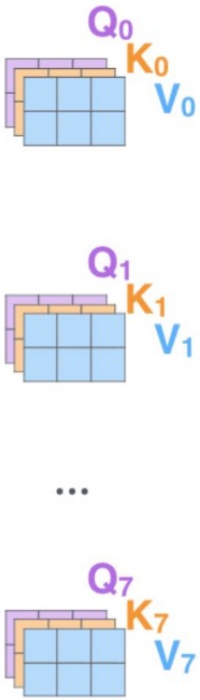


* In all encoders other than #0, we don't need embedding. We start directly with the output of the encoder right below this one

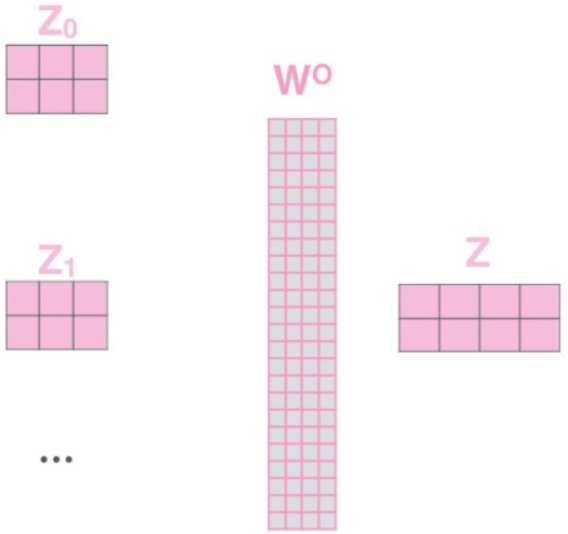
3) Split into 8 heads. We multiply X or R with weight matrices



4) Calculate attention using the resulting Q/K/V matrices



5) Concatenate the resulting Z matrices, then multiply with weight matrix W^O to produce the output of the layer



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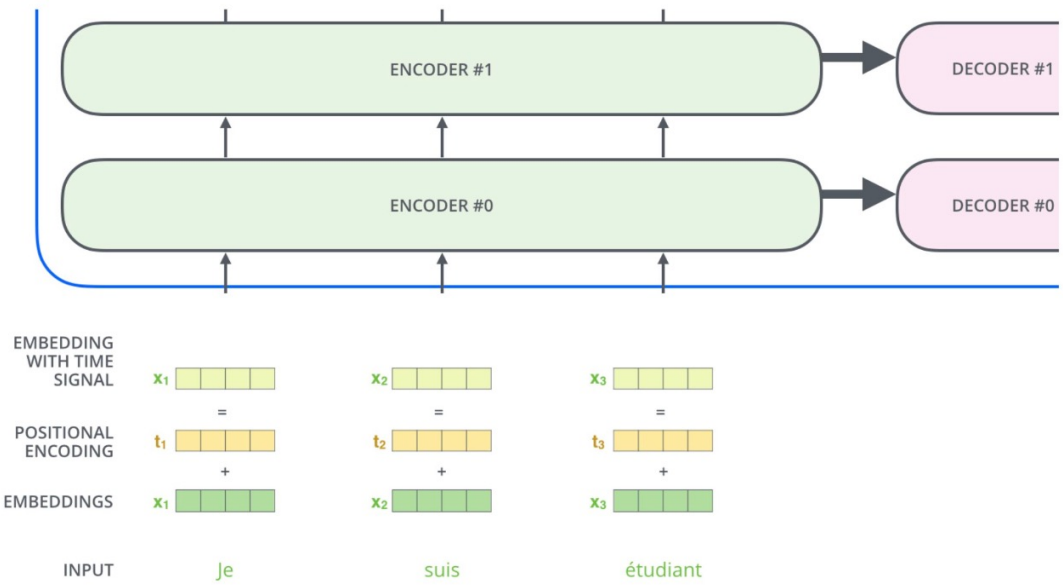
- What is missing so far …?

Another ornament: Positional encodings



- Transformers lack a built-in notion of word order.
- To encode position information, we **add a positional vector** to each input embedding.
- In the original Transformer, fixed sinusoidal positional encodings were used (i.e. sinusoidal functions of different frequencies)

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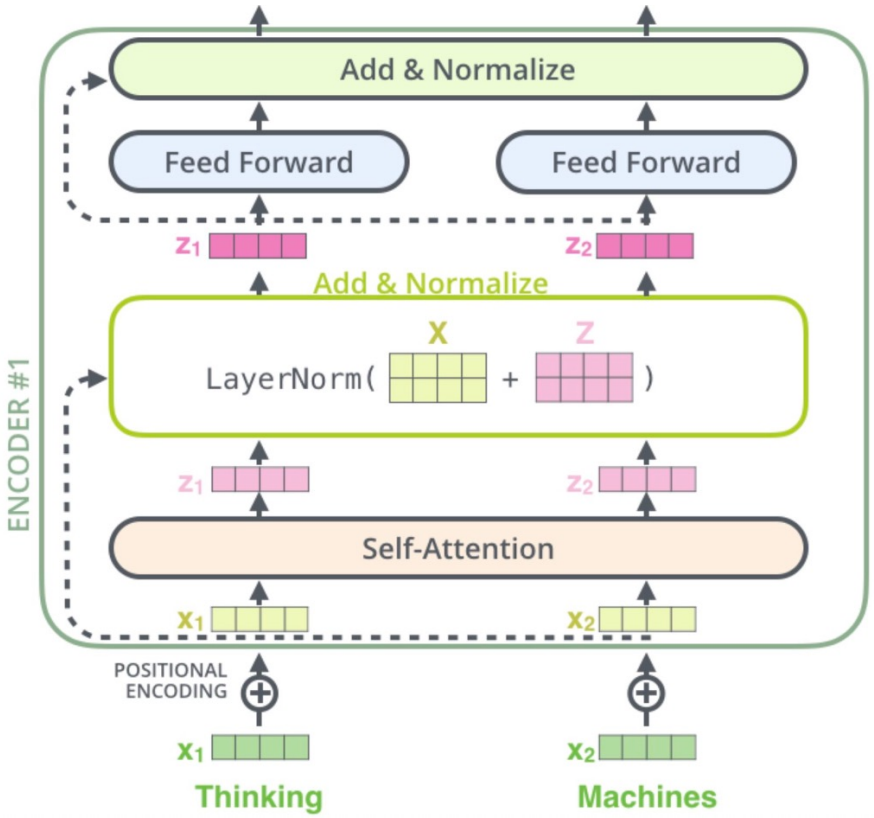
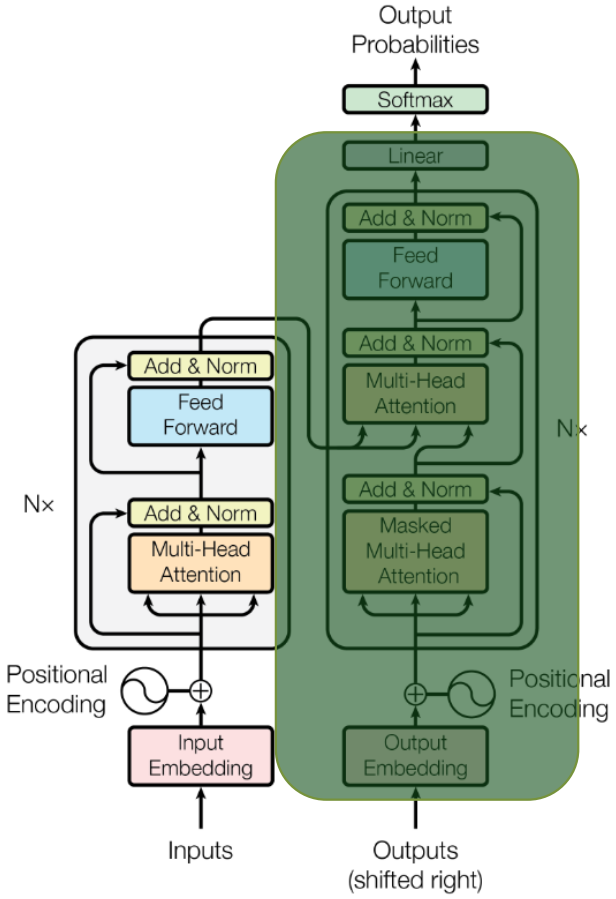
Another ornament: Residuals



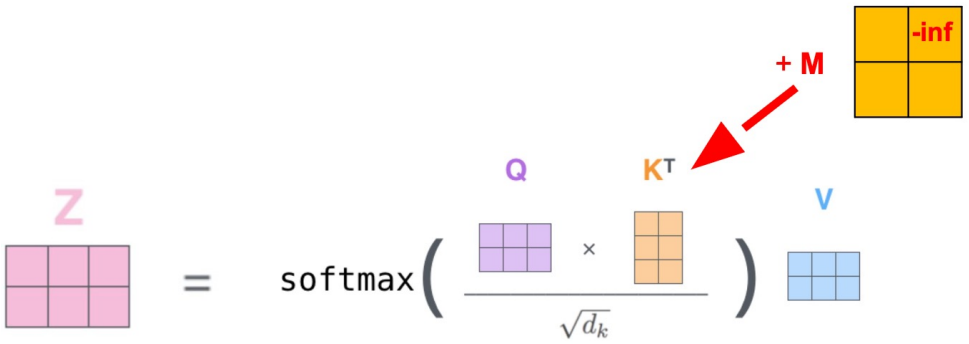
- To stabilize training in deep networks, each sub-layer is wrapped with a residual connection.
- In each sub-layer (self-attention or FFN), the input is added to the sub-layer output.
- The output is then passed through a layer normalization step.
- Residual connections improve gradient flow and facilitate optimization in deep architectures.

The Decoding Process



- **Cross Attention**: allows the decoder to attend to encoder outputs when generating each token
- **Output Generation**: each decoder layer combines the following to produce representations used to predict the next token
  - masked self-attention
  - encoder–decoder (cross) attention
  - a feed-forward network
- **Masked Self-Attention**: ensures each position can only attend to previously generated tokens (prevents access to future tokens)

Masking in the Decoder



- The decoder's self-attention can only look at earlier tokens (i.e. not future ones).
- This prevents information leakage from future tokens during training.
- Implemented by masking future positions, setting their attention scores to -inf before softmax.

Generate token based on previous essentially

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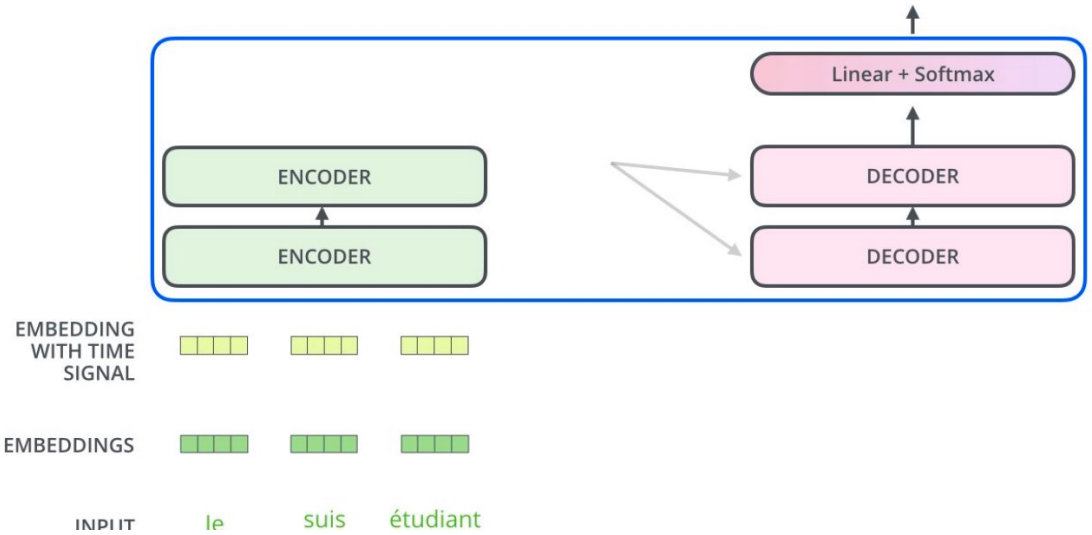
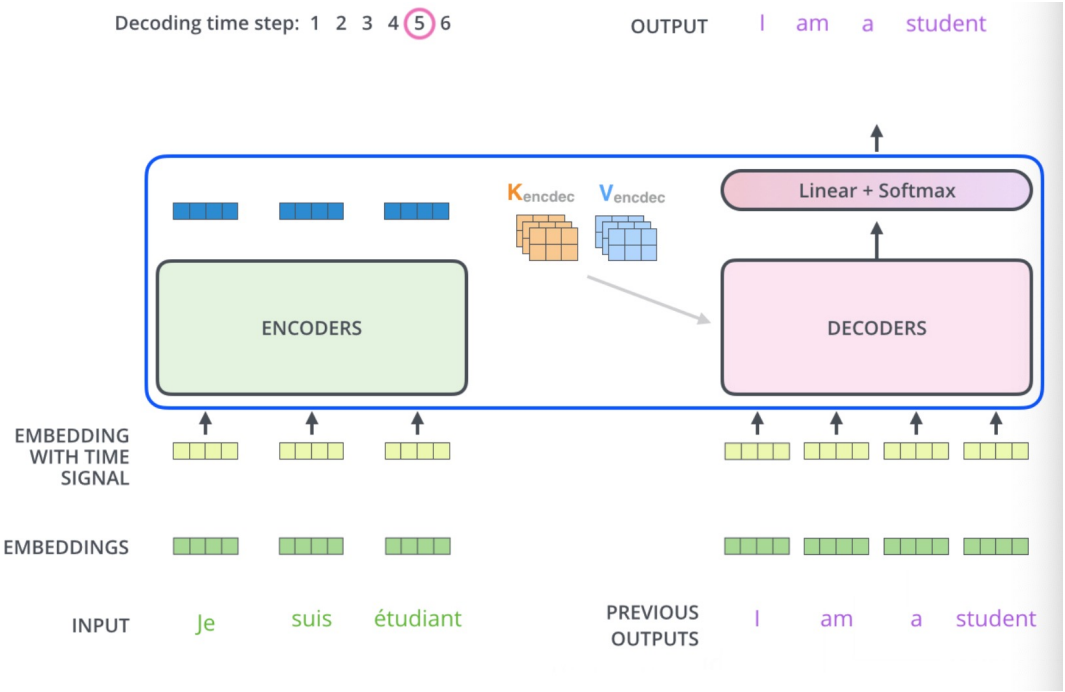
Cross Attention



- The decoder attends to the encoder's output representations to generate predictions.
- The top encoder layer produces contextualized representations (used as **keys** and **values** in cross-attention).
- These are fed into the decoder's **cross-attention** layer.
- This helps the decoder **focus on relevant input positions** during generation.

From Decoder to Output Token



Use teacher-forcing to reinforce generating the next-token.

- **Training time:** the full target sequence is known. We use *teacher forcing*, feeding the correct previous tokens, to compute next-token predictions.
- **Inference time: o**utput is generated autoregressively, *one token at a time*. Each predicted token is fed back into the decoder.
- **Linear Layer**: Projects the decoder output to the vocabulary size.
- **Softmax Function**: Converts logits into probability distributions over the vocabulary.
- **Token Selection**: selects the token (e.g., via greedy decoding or sampling).

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How Transformers Learn: the Training Process

- **Loss Function**: cross-entropy loss is used to measure how far the predicted token probabilities are from the true target tokens. Lower loss indicates better predictions.
- **Backpropagation**: gradients of the loss are computed and used to update model parameters (e.g., via Adam optimizer).
- **Datasets**: trained on paired input–output sequences (e.g., source and target sentences for translation).

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Summary

- **Core Idea**: Overcomes limitations of RNNs and LSTMs (e.g., sequential bottlenecks, long-range dependencies) through global self-attention.
- **Key Components**: Self-Attention, Multi-Head Attention, Encoder-Decoder Structure, Positional Encoding
- **Why It Works**:
  – Short path length between tokens (i.e. $O(1)$ per layer)
  – Fully parallelizable across sequence positions
  – Scales effectively to very large models and datasets

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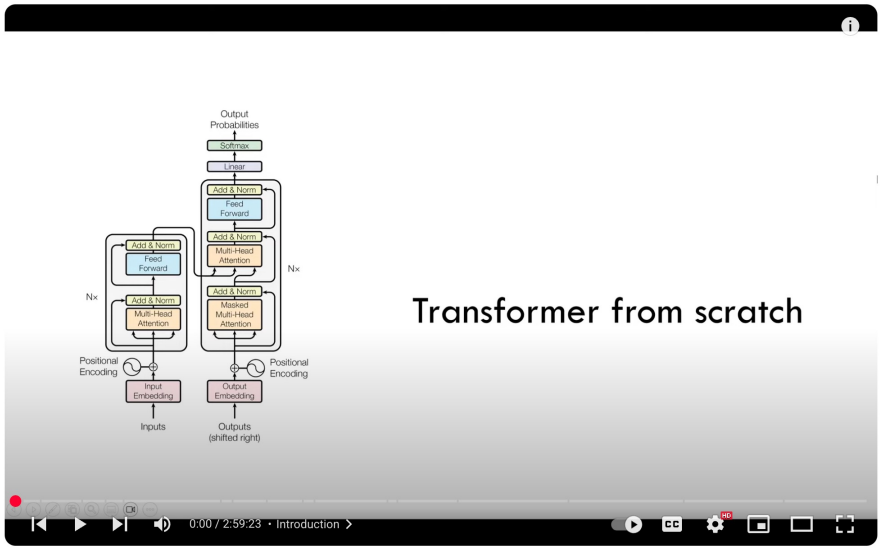
Building Transformers from Scratch



Coding a Transformer from scratch on PyTorch, with full explanation, training and inference.

Umar Jamil
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